

Disaster Spread Ranges & Drone Coverage

A consolidated reference report: flood, landslide, and earthquake spread distances; the Bhopal gas disaster; and drone altitude-coverage relationships

1. Floods

1.1 Assam Floods, 2026 — Initial Stage vs Peak Stage

- East-west spread: approximately 265 km
- North-south spread: approximately 126 km
- Maximum straight-line spread (diagonal): approximately 290 km

This distance figure is a calculation derived from the district list in the reports; the reports themselves do not state a linear spread distance.

Source: Sphere India, "Situation Report 2 – Flood Situation in Assam", 18 July 2026. IFRC, "DREF Operation – Assam Floods 2026" (Appeal MDRIN029), published 6 August 2026.

1.2 Uttarakhand Flash Flood, 2025

- Spread: approximately 9 km, calculated from the coordinates of the outermost affected villages (Dharali and Sukhi Top)
- Origin elevation: approximately 1,550–1,700 m in the surrounding hills

Source: Wikipedia, "2025 Uttarakhand flash flood".

2. Landslides

2.1 Nepal Landslide & Debris Flow, 26 August 2026

- Initial landslide source zone: ~2,000 m long; drop from 5,140 m to 3,990 m elevation
- Core debris flow / flood travel distance: ~100 km downstream
- Extended reach (bodies and debris carried further): up to ~170 km

Source: USGS, "2026 Nepal Debris Avalanche and Flash Flood" event page. European Space Agency, "Nepal flash flood imaged by satellites." Al Jazeera, "Satellite images show destruction from Nepal-Tibet floods." Dave Petley, "The landslide that caused the 26 August 2026 debris flow in Nepal and Tibet," The Landslide Blog, Eos/American Geophysical Union, 7 September 2026.

2.2 Wayanad Landslide, Kerala, 30 July 2024

- Core debris flow distance: 6–7 km down the valley (a typical landslide's debris flow is only 1–1.5 km, making this 4–5 times larger than normal)
- Ground search-and-rescue zone: 6 km stretch
- Extended river search zone (bodies carried downstream via the Chaliyar river): 40 km stretch

Source: Institute of Land and Disaster Management, Department of Revenue, Government of Kerala, "Report on Wayanad Landslide 2024" (ISBN 978-81-960216-4-1), published 23 April 2025. Humanitarian Aid International / LOCAL Platform, "Situation Report – Landslide in Wayanad, Kerala" (Sitrep 011/2024), 5 August 2024.

3. Earthquakes

Unlike floods and landslides, an earthquake has one fixed origin point (the epicenter), and its fault rupture length can be estimated from its magnitude using an established scientific formula (Wells & Coppersmith, 1994): rupture length grows exponentially with magnitude, roughly following $\text{rupture length} \propto 10^{(0.69 \times \text{magnitude})}$.

3.1 1934 Bihar–Nepal Earthquake

- Estimated fault rupture length: ~250–300 km (estimated from magnitude using the Wells & Coppersmith formula, cross-checked against the 1872 Owens Valley earthquake, also magnitude 8.0, which had a recorded rupture of 292 km)

Source: Wikipedia, "1934 Nepal–India earthquake." Wells, D.L. and Coppersmith, K.J., "New Empirical Relationships among Magnitude, Rupture Length, Rupture Width, Rupture Area, and Surface Displacement," Bulletin of the Seismological Society of America, 1994.

3.2 2005 Kashmir Earthquake

- Estimated fault rupture length: ~130–150 km (estimated from magnitude using the Wells & Coppersmith formula)

Source: Wikipedia, "2005 Kashmir earthquake."

4. Bhopal Gas Disaster, 2–3 December 1984

The Bhopal gas leak spread differently from a flood, landslide, or earthquake: instead of following a river valley or a fault line, the toxic gas cloud spread according to wind direction, forming an irregular, unevenly shaped plume across the city rather than a clean circle or line.

- The gas reached Bhopal Railway Station, just 1 km from the factory
- Affected area: estimates of the toxic gas cloud's spread range from approximately 40 km² to 65 km² around the plant
- The report's own retrospective conclusion: if the factory had been sited 3–4 km away from the nearest residential area, most of the deaths would have been prevented — an implicit acknowledgment that the lethal zone extended beyond that distance

Source: Indian Council of Medical Research (ICMR), Bhopal Gas Disaster Research Centre, Gandhi Medical College, "Health Effects of the Toxic Gas Leak from the Union Carbide Methyl Isocyanate Plant in Bhopal: Population Based Long Term Epidemiological Studies (1985–1994)," Technical Report. C&EN/ACS, "40 Years, Still No Closure" and "40 Years Later, Bhopal Is Still in Crisis" (~40 km² estimate). EBSCO Research Starters, "Bhopal Disaster" (~65 km² estimate).

5. Drone Coverage: Altitude, Area, and Recent Technology

5.1 Consumer / Commercial Drones

Drone class	Typical altitude	Ground area covered (per flight)
Legal ceiling (most countries)	120 m (400 ft)	Governs nearly all civilian drone flights
Mid-range consumer drone (e.g. DJI Mavic 3 class)	up to ~6,000 m (manufacturer limit); flown at 100–200 m in practice	~0.12–0.40+ km ² per battery charge (survey use)
Professional survey drone (fixed-wing/VTOL, e.g. WingtraOne class)	300–500 ft typical	Up to ~5.5 km ² in one flight

5.2 Military-Grade Drones

Drone / class	Altitude	Area
Typical military reconnaissance UAVs (e.g. Global Hawk class)	~15,000–20,000 m (50,000+ ft)	~730–1,300 km ² footprint
Airbus Zephyr (solar-powered, world record)	23,195 m (76,100 ft)	~1,745 km ² footprint

Area calculated from altitude using the formula in Section 5.4, assuming a representative 84° camera field-of-view and a square footprint (width²).

5.3 Low-Cost / Toy-Grade Drones

Drone class	Realistic altitude	Area
Very cheap toy drone (~₹2,000)	30–100 m, often unstable even there	~0.003–0.032 km ² footprint
Entry-level GPS drone (~₹15,000–30,000, e.g. DJI Mini-class)	Legally capped at 120 m; firmware may allow up to ~500 m	~0.047–0.81 km ² footprint

Area calculated from altitude using the formula in Section 5.4, assuming a representative 84° camera field-of-view and a square footprint (width²).

What has genuinely improved in low-cost drones by 2026 is battery life, GPS accuracy, and stabilisation — not maximum altitude. Reaching very high altitudes still requires the stronger motors and precise sensors found only in expensive, purpose-built drones.

5.4 How Ground Area Is Calculated from Altitude

A drone camera's view spreads outward from the lens in a cone shape, exactly like a torch beam — the higher the drone flies, the wider the visible circle on the ground becomes. This is calculated with a simple formula:

Ground footprint width = $2 \times \text{altitude} \times \tan(\text{camera field-of-view} \div 2)$

Doubling the altitude doubles the footprint's width and length, which means the area covered on the ground becomes four times larger — but the level of detail (resolution) in the image drops by half. This is why disaster-response drones typically fly high for a first wide hazard-scan, then descend low for close, detailed searching.

Source: Formula confirmed in: "Optimising observing strategies for monitoring animals using drone-mounted thermal infrared cameras," arXiv:1812.05498; and in U.S. Patent documents on aerial vehicle map determination (e.g. US Patent 12282989).

5.5 Recent Technology: Multi-Camera Stitching

A drone's altitude is limited by legal rules and by how much detail is needed in the image, so simply flying higher is not always an option for covering more ground. A more recent, low-cost alternative is to mount several small, ordinary cameras on the same drone, each pointed at a slightly different patch of ground, and use software to stitch their images together into one larger combined picture — without needing to fly any higher at all.

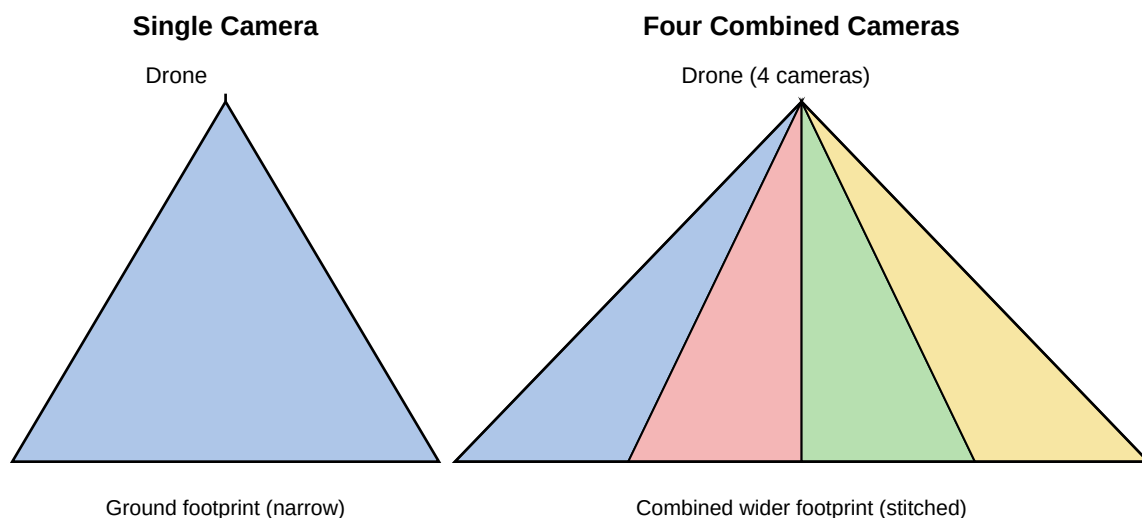


Illustration: a single camera's narrow footprint (left) compared with four cameras combined into one wider, stitched footprint (right).

This method has been demonstrated using real, working hardware: a four-camera rig, tested for photogrammetric survey and 3D building reconstruction, in which the four overlapping images were rectified, matched, and merged into one larger combined image. Similar multi-camera rigs used in other published research use ordinary action cameras (such as GoPro-style cameras) rather than expensive equipment, keeping the approach genuinely low-cost. The same principle — two ultra-wide lenses stitched into one image — is already used in inexpensive, commercially

sold 360° consumer drones today.

Source: "Mini-Unmanned Aerial Vehicle-Based Remote Sensing: Techniques, Applications, and Prospects," *arXiv:1812.07770* (Figure 9, four-combined camera system). "Multi-Camera Imaging System for UAV Photogrammetry," *PMC/ResearchGate*. UAV Coach, "360 Drones: Top Models on the Market & More," 2026.